DSC-002: Mathematical Methods

Lecture 1: *Linear Algebra Prelims*

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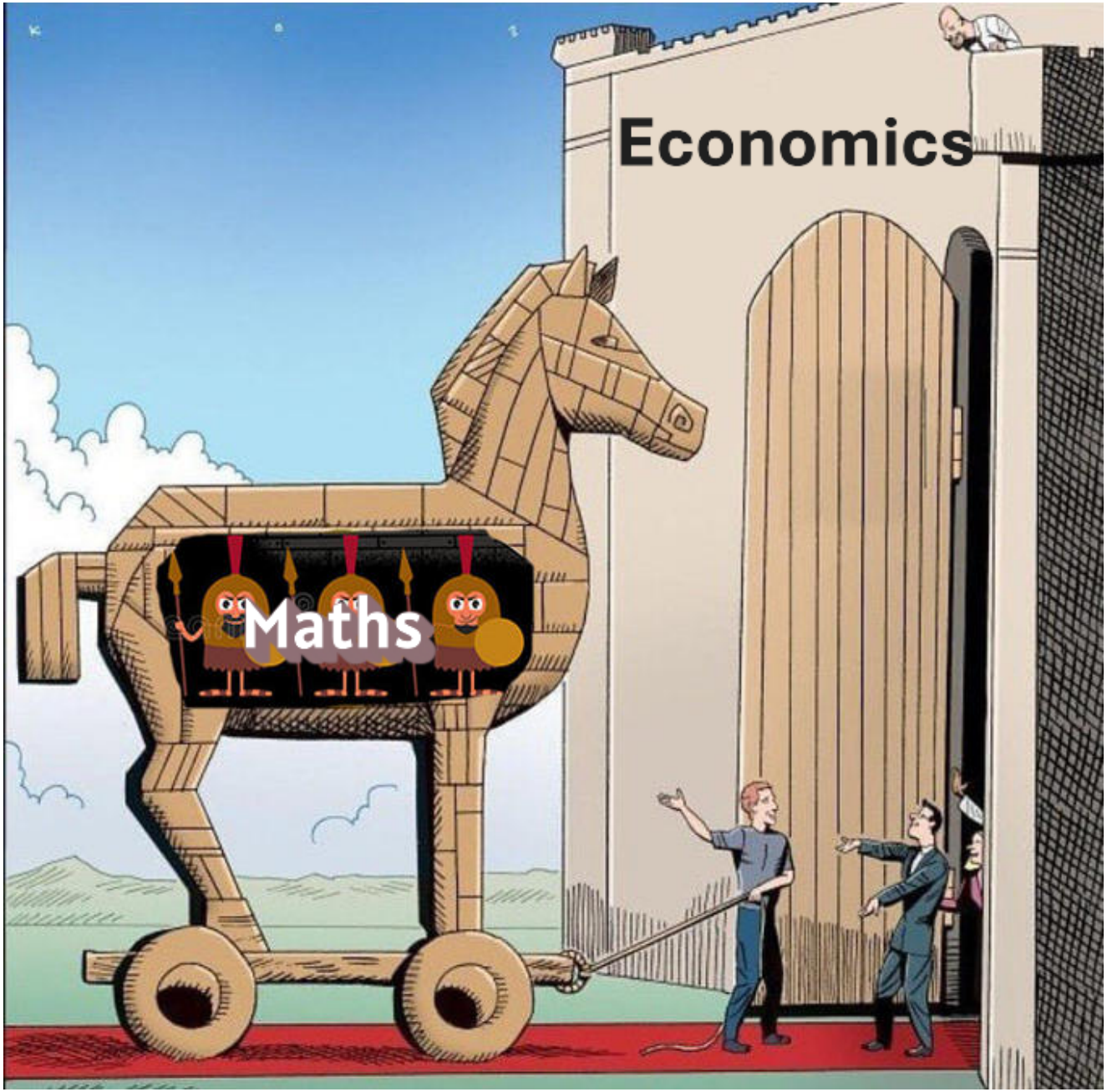
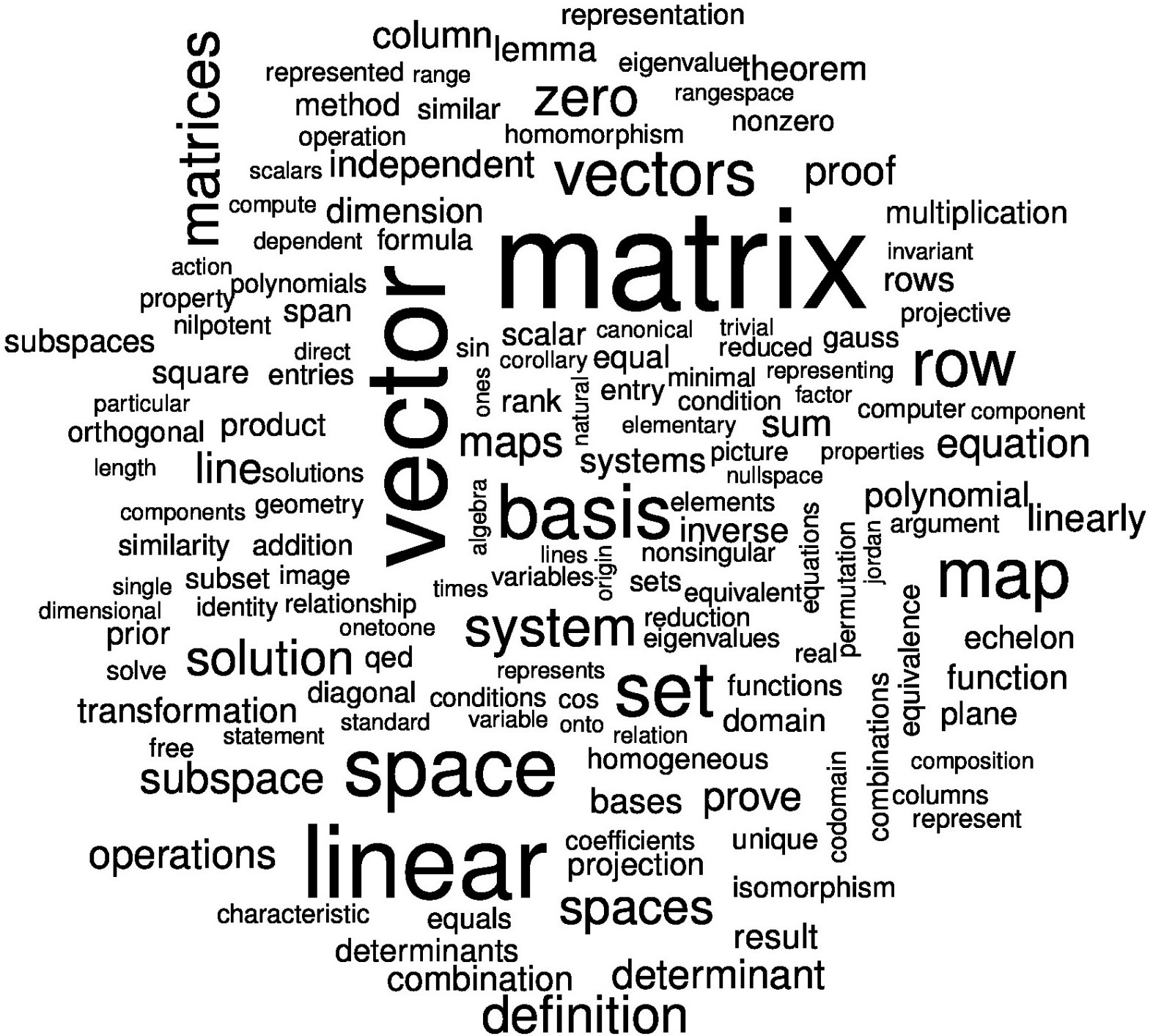
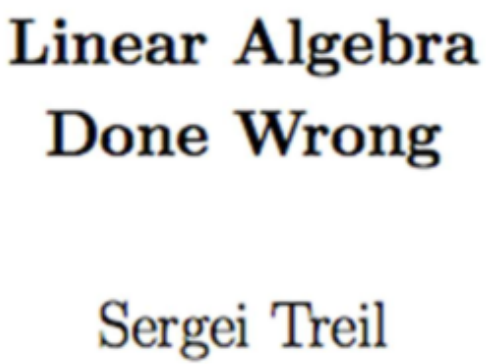


Figure: You wanted study Economics? Okay!!

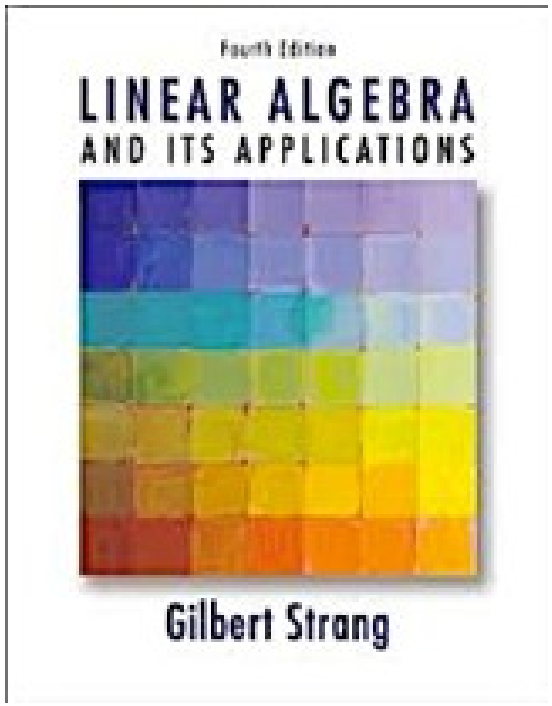

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Recommended Textbooks

Linear Algebra Done Wrong
Sergei Treil (Ed. 2024)



Linear Algebra and Its Applications
Gilbert Strang (4th Ed)



Download PDF (Treil 2024)
(Official free version from author's website)

Course Outline

Trei 2024	Strang 4 Ed.
Lec_slide-3: Foundation of LA	
Ch 1: **Basic notions** Sections: 1.1, 1.2, 1.3, 1.4, 1.5, 1.5, 1.6, 1.7 (pages: 1- 30)	Ch 1: Sections: 1.1, 1.2 Ch 2: Sections – 2.1, 2.2, 2.3, 2.4, 2.6
Ch 2: **Systems of linear equations** Sec: 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8	Ch 1: Sections – 1.3,
Lec_slide-4: Inner Product Spaces	
Ch 5: **Inner product spaces** Sections: 5.1, 5.2, 5.3, 5.4, 5.5	Ch 3: Sections – 3.1, 3.2, 3.3, 3.4,
Lec_slide-5: Spectral Theory	
Ch 4: **Eigenvalues and eigenvectors** Sections: 4.1, 4.2	Chapter 5: Sections – 5.1, 5.2,
Lec_slide-6: Quadratic Forms	
Ch 7: **Bilinear and quadratic forms** Sections: 7.1, 7.2, 7.3, 7.4	Ch 6: Sections – 6.1, 6.2
For your reference but NOT covered through lectures	
Chapter 3: Determinants, Sections 1-6; → For your own reference	A handout on Complex Numbers (for those who need to revise)

What's Linear algebra? Basic Concepts

- Linear algebra is the branch of mathematics concerning *vector spaces* and linear mappings between such spaces.
- It includes the study of *lines*, *planes*, and *subspaces*; also involves the analysis of *rotations in space*, *least square fitting*, solution of coupled differential equations, determination of a circle passing through three given points; but is also concerned with properties common to all vector spaces.
- Linear algebra is our ticket to *multidimensional space*.
- Unavoidable if we are into economics, statistics, econometrics, computer graphics, physics, chemistry or anything quantitative. In today's world, that's almost everything!!

- We will start with Vector Spaces in a while, but before that....

What is "Linear" in Linear Algebra?

- **"Algebra"** means, roughly, **"relationships"**.
- School algebra explores the relationship between unknown numbers. Without knowing x & y, we can still work out that expression

$$(x+y)^2 = x^2 + 2xy + y^2.$$

But this is a quadratic function. [Why?].
Finding the roots of a polynomial is a nonlinear problem.

- "**Linear Algebra**" means, roughly, "*line-like relationships*".

What is "Linear" in Linear Algebra? ...

- The word "*linear*" is to be understood as in a linear function in calculus [i.e. a line], such as

  $$y = g(x) = m * x, \qquad \text{where } m, y \ \& \ x \text{ are numbers.}$$

  $\longrightarrow$ Said to be a linear relation (*through origin*) between y & x.

  Or, add a constant, $\qquad y = f(x) = m * x + a,$

  $\longrightarrow$However, this one is **"affine"** *function* [*from the Latin, affinis, "connected with"*], i.e. still a straight line, but not through origin, has an intercept term.

- More generally, if $\mathbf{a}_i$ be the coefficients (fixed *real numbers)* to the $\mathbf{x}_i$ variables and $\mathbf{c}$ be the constant, then a linear equation be -

  $$a_1x_1 + a_2x_2 + \cdots + a_nx_n = c$$

Non-linear case [1]: Illustration & Issues

- **Ex:** Let $\Re$ denote the real numbers, and suppose we want to know something about the set of common solutions of the equations
$$z = x^2 + xy^5, \text{ and}$$
$$z^2 = x + y^4$$
- These equations represent *two* **surfaces** in *real three space* $\Re^3$(why?), so we'd expect the set of common solutions to lie on a curve.
- *Nonlinear systems*, as in the above example, are usually difficult to solve; their theory involves highly sophisticated mathematics
- Here it's impossible to express the solutions in a "**closed form**" , but we can study them "**locally**" using linear methods.
- For example, both surfaces meet at $(0, 0, 0)$ [any other point??], and they both have a tangent plane at $(0, 0, 0)$. The tangent line to the *curve of intersection* at $(0, 0, 0)$ is the intersection of these two tangents.

Non-linear case [2]: Linear Approximations

- The Taylor's expansion of a function f in calculus [you know this!?] is -

$$f(x) = f(x_0) + f'(x_0) * (x - x_0) + 1/2! * f''(x_0)(x - x_0)^2 + \ldots$$

- If we use only the *first two terms* of the Taylor expansion of f, we have a linear approximation to the above function f:

$$\begin{aligned} f(x) &\approx f(x_0) + f'(x_0) * (x - x_0) \\ &= f'(x_0) * x + [f(x_0) - f'(x_0) * x_0] \end{aligned}$$

- If we set

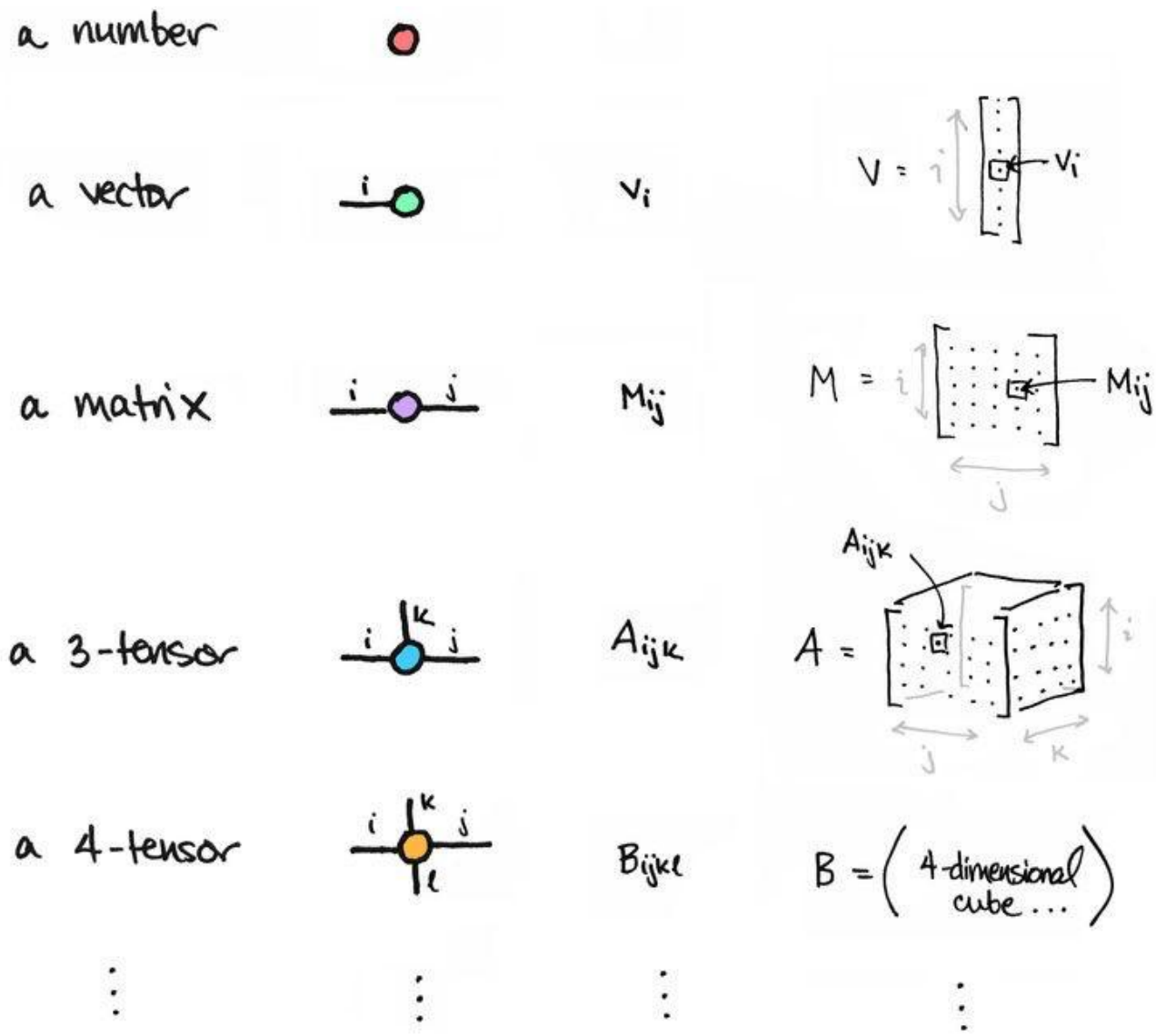
$$m = f'(x_0), \quad \text{and} \quad a = [f(x_0) - f'(x_0) * x_0],$$

- We can see that this is a linear function just like any **affine** function $f(x)$ above. This approximation works reasonably well near x_0
- It is the simplest approximation to f beyond saying "*near* x_0, *it's probably something like* $f(x_0)$."
  That's why the linear approximation is called easy".

Non-linear case [3]: How linear algebra plays a role

- Now what do we do if we have a function that depends on several variables (or, which is the same, a variable x that is a *"vector"* containing these variables as components)? This is in the domain of multivariate calculus (because, multiple variables).
- In that case, $f'(x_0)[:= \mathbf{m}]$ becomes a *vector* that has to be multiplied [how?#] with anther vector $(x - x_0)[:= \mathbf{a}]$.
- Likewise, $f''(x_0)$ becomes a matrix (actually, a *bilinear tensor*) that has to be multiplied twice with the vector $(x - x_0)$. [how? Will see later]
- As we have seen, the linear and quadratic approximations are the easiest ways to think about complicated functions f.
- Linear algebra deals with vectors, matrices, and tensors, and how to operate them with each other so that everything makes sense and is useful too. Let's delve into it!

Illustration – Vector, Matrix, Tensor



System of Simultaneous Equations & it's Solutions

Let there be a function $f : \Re^n \to \Re^m$.

Fix $y \in \Re^m$. Then the equation $f(x) = y$ represents a system of 'm' *simultaneous* equations in 'n' variables.

The set of equations is -

$$\begin{array}{c} f_1(x_1, \cdots, x_n) = y_1 \\ \vdots \quad \ddots \quad \vdots \quad \vdots \\ f_m(x_1, \cdots, x_n) = y_m \end{array} \longmapsto \text{say}, \quad \begin{array}{c} a_{11}x_1 + ... + a_{1n}x_n = y_1 \\ \vdots \ddots \vdots \ddots \vdots \quad \vdots \\ a_{m1}x_1 + ... + a_{mn}x_n = y_m \end{array}$$

where,

$\mathbf{y} = (y_1, ..., y_m)^T$, $\mathbf{x} = (x_1, ..., x_n)^T$, (bold faces letters are col. vectors)

$f(x) = (f_1(x), ..., f_m(x)), \ \forall x \in \Re^n$.

The *component function* $f_i : \Re^n \to \Re \ \forall \ i \in \{1, ..., m\}$.

System of Simultaneous Equations & it's Solutions...2

$$A = \begin{bmatrix} a_{11} \cdots a_{1n} \\ \vdots \ \ddots \ \vdots \\ a_{m1} \cdots a_{mn} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} & \cdots & \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \end{bmatrix}$$

where each column $\mathbf{a}_k = \begin{pmatrix} a_{1k} \\ \vdots \\ a_{mk} \end{pmatrix}$ represents the kth **column vector**.

- *vector* $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ that satisfies $f(x) = y$ is a **solution** to system.
- The system in matrix notation is: $A_{m\times n}\mathbf{x}_{n\times 1} = \mathbf{y}_{m\times 1}$, and
  . solution is $\mathbf{x} = A^{-1}\mathbf{y}$, using a matrix inverse.
- **Note that** whether the system of equations has a solution is the same question as whether $f^{-1}(\{\mathbf{y}\})$ is a "*nonempty*" set.
- Other techniques e.g. Gaussian elimination are numerically more robust.

Simultaneous Equations - 'Contour surfaces'

Example

Consider a function $f : \Re^2 \to \Re^2$, where $f(x_1, x_2) = (x_1 x_2; \quad x_1 + x_2)$.
For what values of y in the codomain does the equation $f(x) = y$ have a solution?

[Let's *draw an analogy with consumer's equilibrium problem*]

One way to look at a solution: the intersection of (hyper) surfaces.

For the example above, given $y = (y_1, y_2)$, there are two equations:

$$x_1 x_2 = y_1 \qquad \text{and} \qquad x_1 + x_2 = y_2.$$

Both of these are **1-dimensional surfaces / curves in** $\Re^2$, and their intersection is the set of solutions.

For the more general n-variable & m-equation case, $f_i(x_1, ..., x_n) = y_i$ describes an $(n-1)$-**dimensional surface in** $\Re^n$, and the solution set is the intersection of the 'm' such surfaces.

System of Linear Equations & it's Solutions ... [1]

- **Example-1:** - *Basic Idea*

$$x - 2y = 1 \quad (1)$$
$$3x + 2y = 11 \quad (2)$$

- In matrix form -

$$\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix},$$

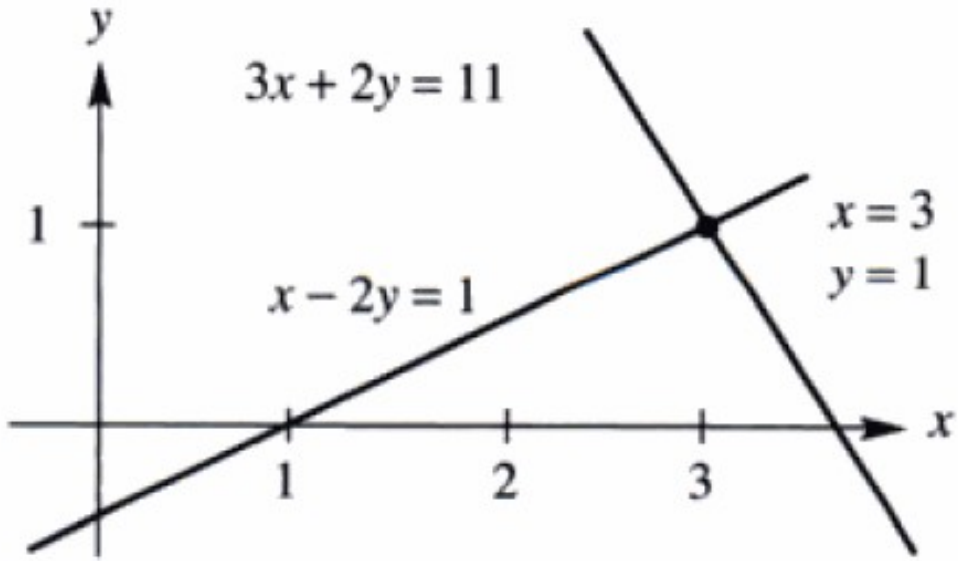
[what are the vectors here?]

$$\Rightarrow \quad [A]\overrightarrow{X} = \overrightarrow{b} \qquad \Rightarrow \overrightarrow{X} = [A]^{-1}\overrightarrow{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

- The '*ordered pair*' $[3, 1]^T$ is the solution, and not $[1, 3]^T$
- What about row reduction method? TRY IT!!

System of Linear Equations & it's Solution ... [2]

- (Ex-1) **Row (Line) Picture**:
  Let's draw the conventional picture: one equation is one line.

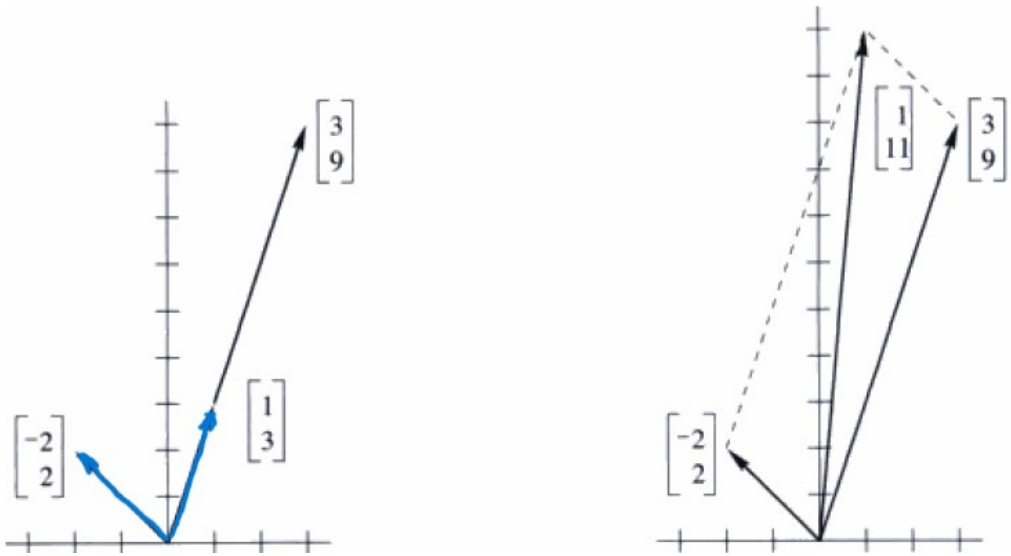


- Solution is the intersection between the two. Simple for 2-eqn case!!
- What about if a system has: 3 or more eqns? more than 2 variables?
- The matrix inversion method is often cumbersome, and beyond 3-D graphical method is next to impossible!!

System of Linear Equations & its Solution

- (Ex-1) **Column (Vector) Picture**: Try to recognize the linear system as vector form eqn. Separating original system into columns instead of rows obtain a "linear combination" of the column vectors of matrix A:

$$x\begin{bmatrix}1\\3\end{bmatrix} + y\begin{bmatrix}-2\\2\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix} = b \quad \Rightarrow \text{Solution: } 3\cdot \text{col}_1 + 1\cdot \text{col}_2 = b$$



- Applying **vector addition** method, we get the solution.
- **Parallelogram Law**: [can you recollect?]

System of Linear Equations & it's Solutions ... [4]

- **Example-2**: with 3 unknowns with 3 equations ...

$$x + 2y + 3z = 6 \quad (1)$$
$$2x + 5y + 2z = 4 \quad (2)$$
$$6x - 3y + z = 2 \quad (3)$$

- In matrix form: $\rightarrow$ Linear combn of column vectors?

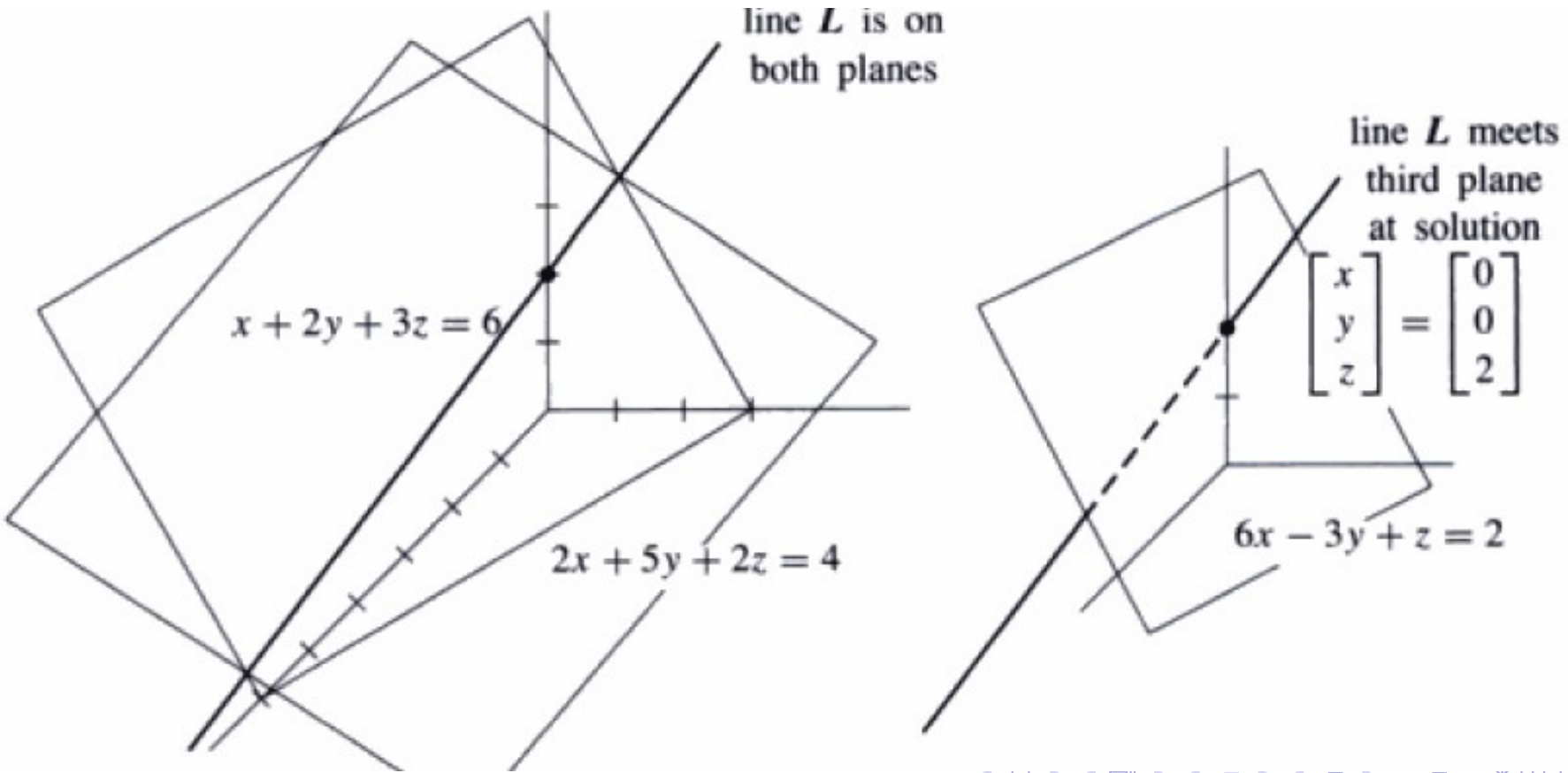
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix},$$

Solution to $[A]\overrightarrow{X} = \overrightarrow{b} \quad \Rightarrow \overrightarrow{X} = [A]^{-1}\overrightarrow{b} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ 2 \end{bmatrix}$

- Can you see some thing interesting here?? (first 2 entries)
- What about row reduction method? TRY IT!!

System of Linear Equations & it's Solution ... [5]

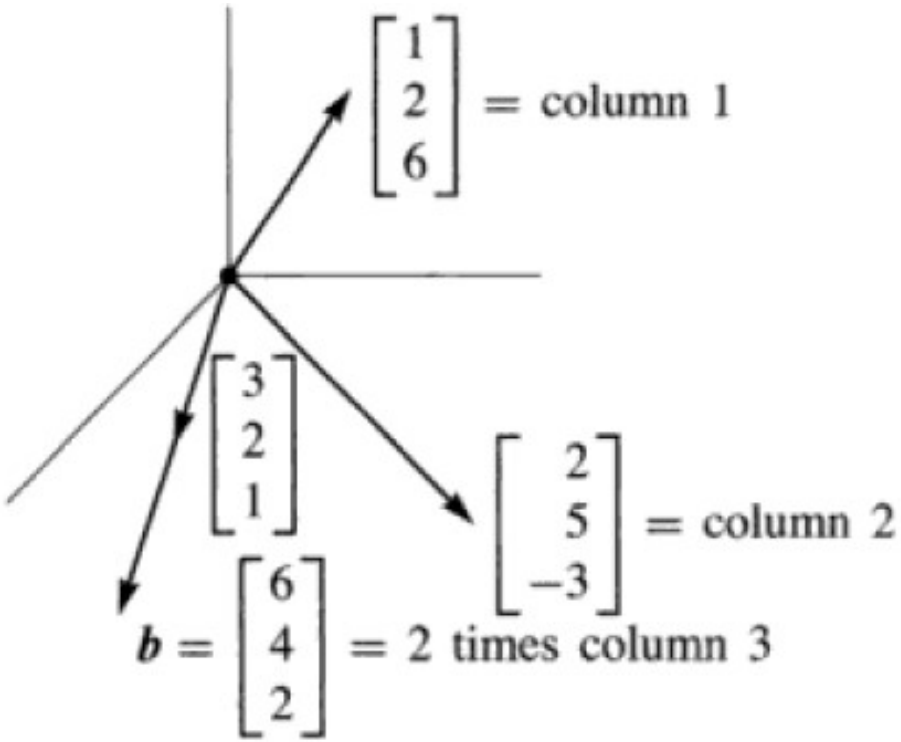
- (Ex-2) **Row (Line) Picture**: Each equation is a plane in 3-D space. First equation crosses X, Y and Z axes at points $(6, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 2)$ respectively, and that define the plane. Similarly others.



System of Linear Equations & it's Solution ... [6]

- (Ex-2) **Column Picture**: Let's start with the vector form of equation

$$x\begin{bmatrix}1\\2\\6\end{bmatrix} + y\begin{bmatrix}2\\5\\-3\end{bmatrix} + z\begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}6\\4\\2\end{bmatrix} \Rightarrow \text{Correct combination is } (0, 0, 2)$$



Simultaneous Equations in Economics.

- $\{$*First order conditions in optimization problems; General equilibrium & Nash equilibrium in some problems*$\}$
- All these can be cast as solutions to a system of equations $F(x) = 0$, where $F : \Re^n \to \Re^n$ and $\overrightarrow{0} \in \Re^n$.
- **General equlm** is often described as a **fixed point of a fn**. (i.e., say $f : \Re^n \to \Re^n$, x is a *fixed point* if it satisfies $f(x) = x$). But x is a fixed point of f *iff* it is a zero of $F(x) \equiv f(x) - x$, so the the question is really of finding the zeros of F i.e. solving $F(x) = 0$.
- **Continuity** of F is an important player in the *existence of a solution*. (Just as in 1-D case: if $f : \Re \to \Re$, is continuous, and $f(x_1) > 0 > f(x_2)$, then the **intermediate value theorem** assures a solution $x \in (x_1, x_2)$.) For the more general higher dimensional case, in computational economics methods like Gauss-Jacobi make use of 1-dimensional solutions to the n different equations in an iterative way to converge to a solution.

Reference:

- Strang 4th Edition:
  1. Chapter 2 (for solution to linear systems)
  2. Chapter 1 (for intro to vector analysis)